Footprint calculations

Lecturer: Dr Amanda Irwin
Unit coordinator: Prof. Arunima Malik
amanda.irwin@sydney.edu.au
arunima.malik@sydney.edu.au

THE UNIVERSITY OF SYDNEY



We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

THE UNIVERSITY OF SYDNEY

Mid-semester survey



Homework 1 submission

Homework Submission 1

Worth: 25% of final mark
Due: 9 September, 2026, 11:59PM

You must submit **two separate files**:

Excel File

Submit under **Assignments** (**Homework Submission 1: Excel File**)

Your Excel workbook should:

- Include completed homework tasks for **Weeks 1–5**.
- Use **a separate worksheet (tab) for each week**.
- Display **all formulae** used in calculations (do not paste values only).
- Clearly label **all data, variables, and units**.

Important:
Your submission **must include the required number of sectors specified in each week's homework task. If your submission does not include the required number of sectors, a penalty of 50% of the maximum mark available for that week's homework will be applied.**

Explanation File (Word or PDF)

Submit under **Assignments** (**Homework Submission 1: Word/PDF File**)

Prepare a Word document containing your answers to the **Explanation** questions for **Weeks 1–5**. You can either submit a Word file or a PDF file.

The Explanation questions are designed to assess your understanding of the methodology underlying input–output calculations.

Important:

- Explanation answers included in the Excel file **will not be marked**.
- Failure to submit the Explanation file will result in **zero marks for the Explanation component** of Homework Assignment 1.
- Explanation **answers that include terminology that is unrelated to the content taught in this unit may have marks deducted**.

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Reminder from week 1

Using generative AI

Generative AI tools, such as Microsoft Copilot Chat, can create digital content, including text, images, and video. Different guidelines for use of generative AI apply to learning and assessment:

Generative AI for general learning

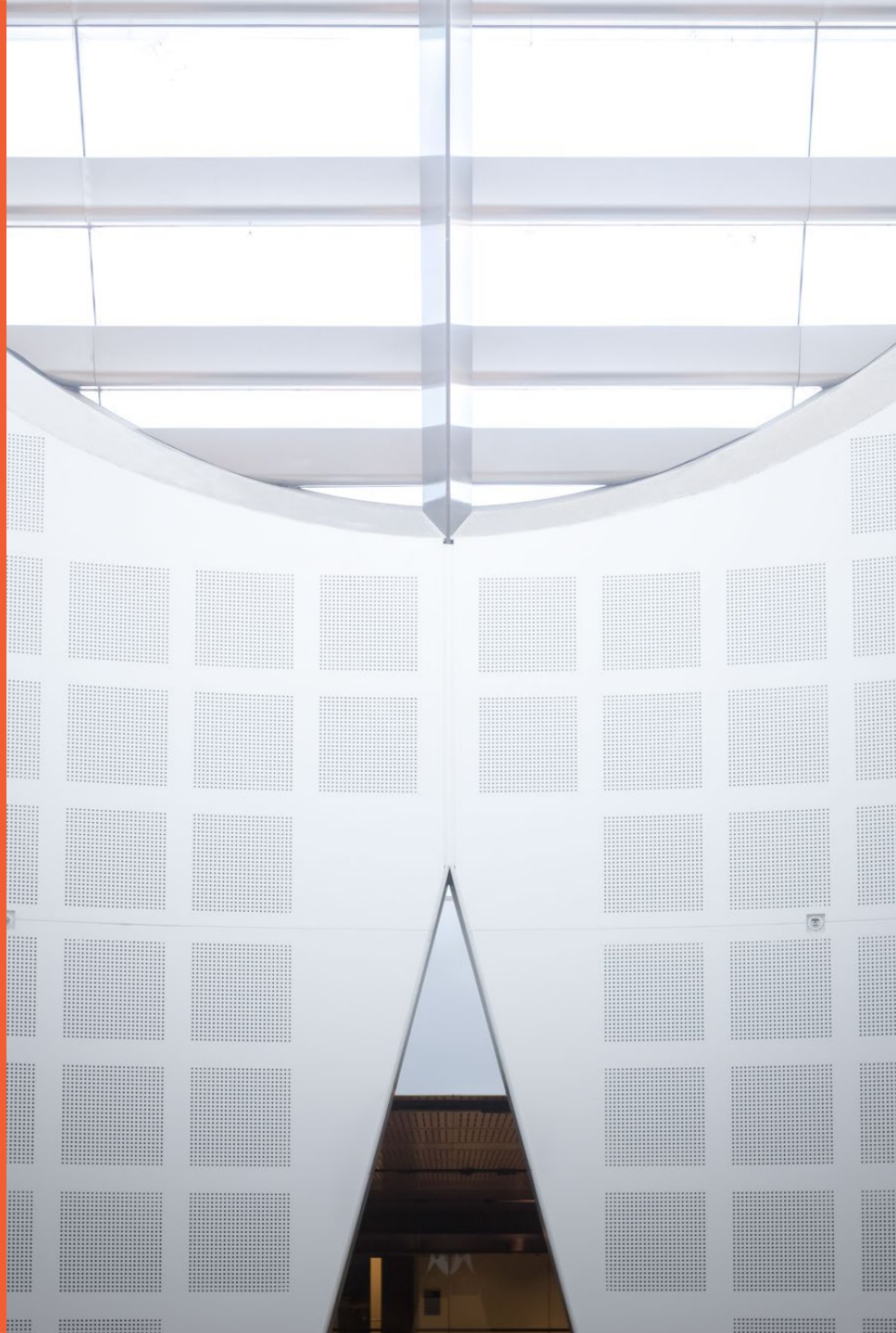
- You can use generative AI to help you learn. For example, it can explain things, apply knowledge, plan your study and make practice questions. Use our student co-designed AI resource so you are aware of its strengths and limitations.

Generative AI for assessment

- Check your unit outline for whether you can use generative AI tools in your assessment:
    - For in person supervised tasks like exams – generally you will not be allowed to use AI unless your coordinator expressly permits it.
    - For other tasks, generally you will be able to use it, but you must appropriately acknowledge all use.
    - Failing to acknowledge AI use is an academic integrity breach.
- When using generative AI tools, don't provide content generated by other students, others' intellectual property (unless with permission), or personal or health information. Your unit coordinator can provide guidance on this.

AI resource made by students:
https://bit.ly/students-ai

PHYS5033

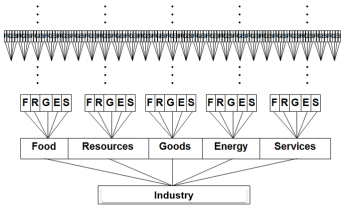


THE UNIVERSITY OF SYDNEY

Recap – Week 5

- We can add a satellite matrix, or **Q** matrix, to an IO table to include social, environmental, or economic indicators
- Potential sources of data to populate the **Q** matrix
- Sector structures used for **Q** matrix data sources and the IO tables we partner them with are often different
- We use a concordance, a proxy, and the mapping steps to convert the data into a sector structure that matches our IO table

PHYS5033 journey – where are we?



WEEK 1

- The need to do sustainability supply chain assessments
- What are footprints?
- How to calculate footprints? Need Input-output tables. Why? IO tables offer completeness in terms of supply chain coverage

WEEK 2
Recap of matrix algebra

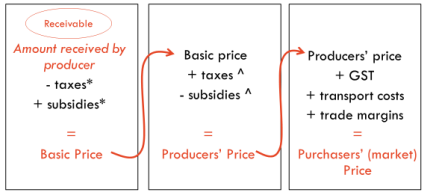
Matrices
Vectors
Multiplication
…

Sustainability assessment



WEEK 3

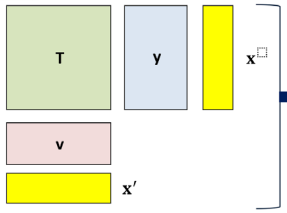
Valuation and different pricing options in the SNA

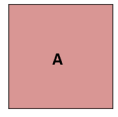


* taxes and subsidies on the production process (eg payroll tax, land tax, jobkeeper)
^ taxes and subsidies on the product itself (eg tobacco & alcohol excises)

WEEK 4

Recap – economic system

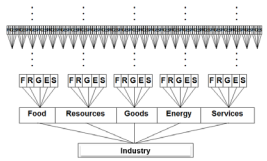




$L = (I - A)^{-1}$

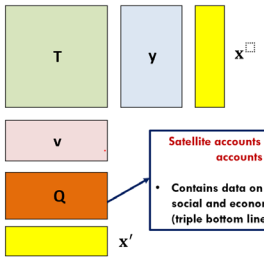
- Create an identity matrix (same size as the **A** matrix)
- Calculate **I** – **A**
- Calculate the inverse of **I** – **A** using minverse

- **T** matrix; **A** matrix; **L** matrix – SAME SIZE!
- **A** matrix – Direct requirements matrix
- **L** matrix – Total requirements matrix



WEEK 5

Recap – physical/ satellite system



Satellite accounts matrix/Physical accounts matrix

- Contains data on environmental, social and economic indicators (triple bottom line)

What should I know by now?

- What a supply chain is
- What an input-output table is, and why we use input-output analysis for measuring supply chain impacts
- The structure of input-output tables
- How to undertake matrix calculations in Excel
- Basics of matrix manipulation
- What the **A** and **L** matrices are, and how to calculate them
- What the fundamental input-output equation is
- How to construct a **Q** matrix using environmental / social / economic indicators

What should I know by now?
Learning outcomes

LO1: Appreciate the national accounts and their role in society

LO2: Brush up your linear algebra and matrix operation skills

LO3: Understand the fundamentals of input-output theory and environmental extensions

LO4: Understand the workings of economic demand-pull model

Next: Bringing the economic and physical systems together

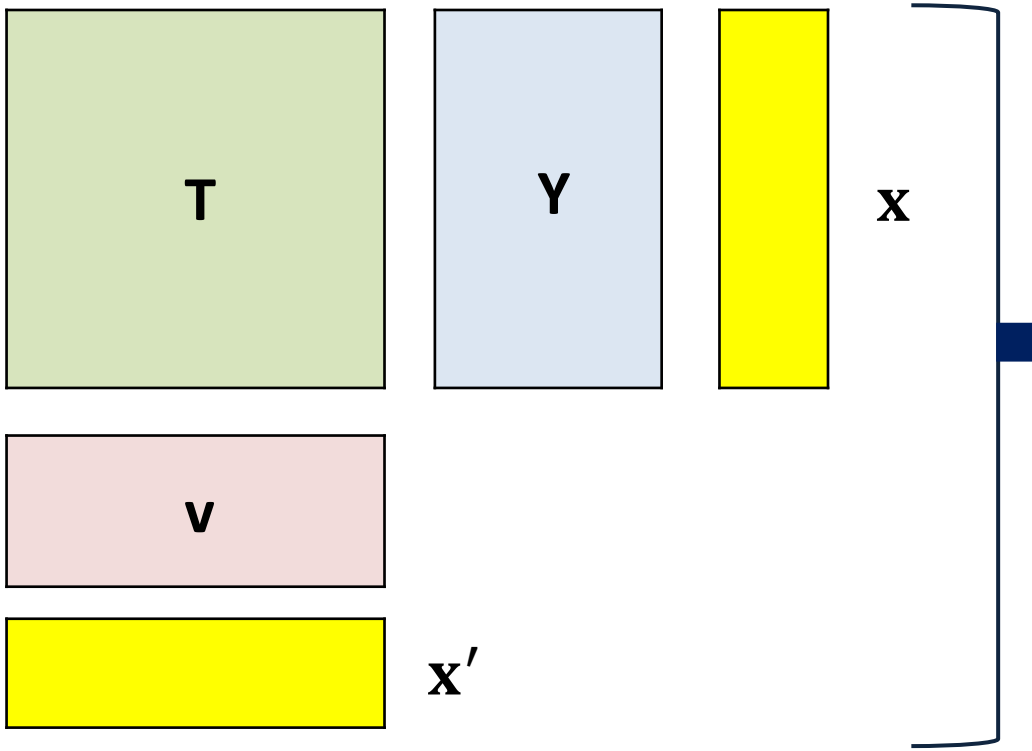
LO5: Understand the applications of input-output analysis to environmental issues

LO6: Deepen your knowledge of multi-region input-output models, applications of footprinting, and MATLAB for sustainability assessments

LO7: Apply input-output analysis to applications from a range of disciplines

LO8: Apply input-output analysis to business settings

Recap – economic system



$$\mathbf{A} = \mathbf{T}\hat{\mathbf{X}}^{-1}$$

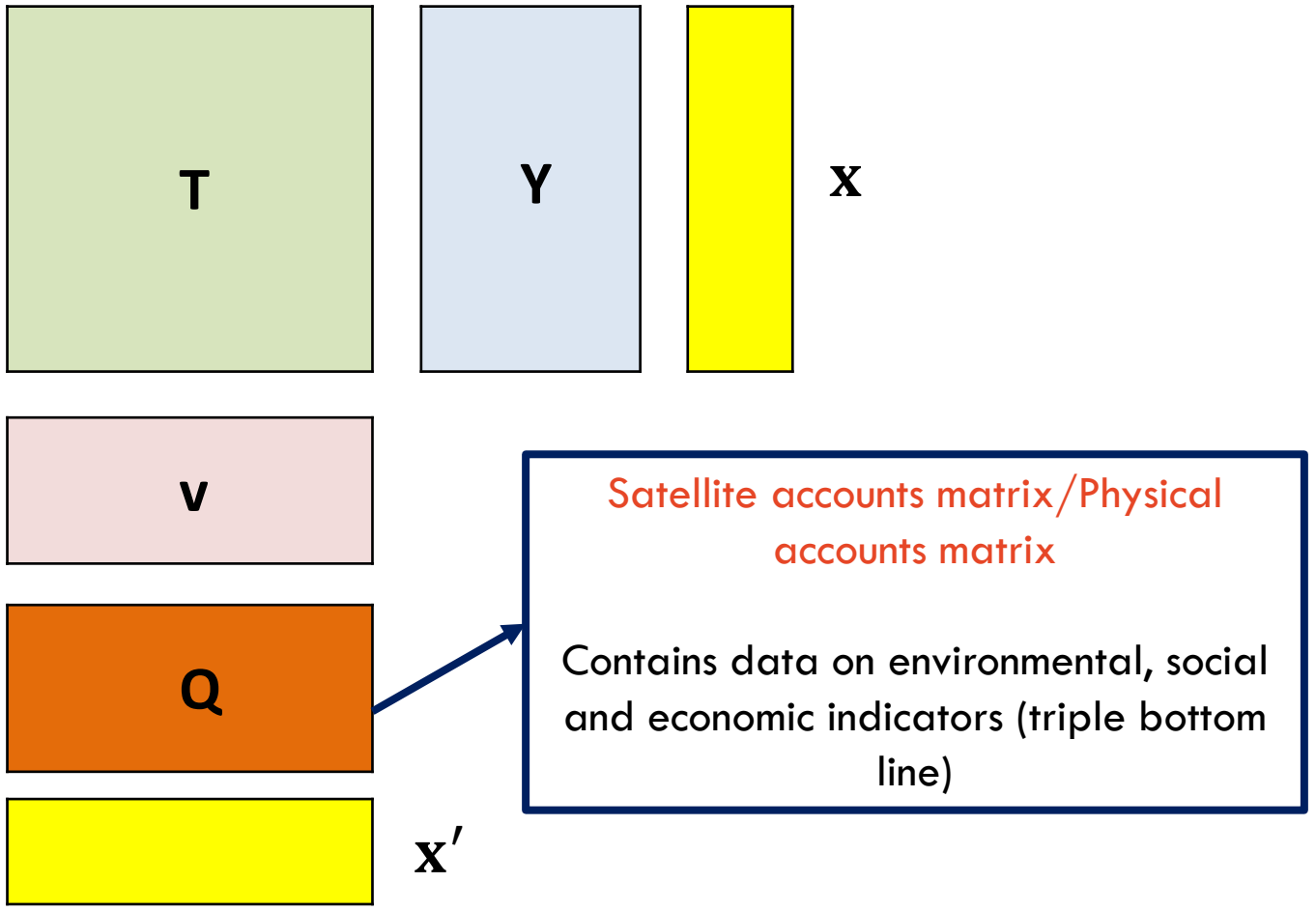
A

$$\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$$

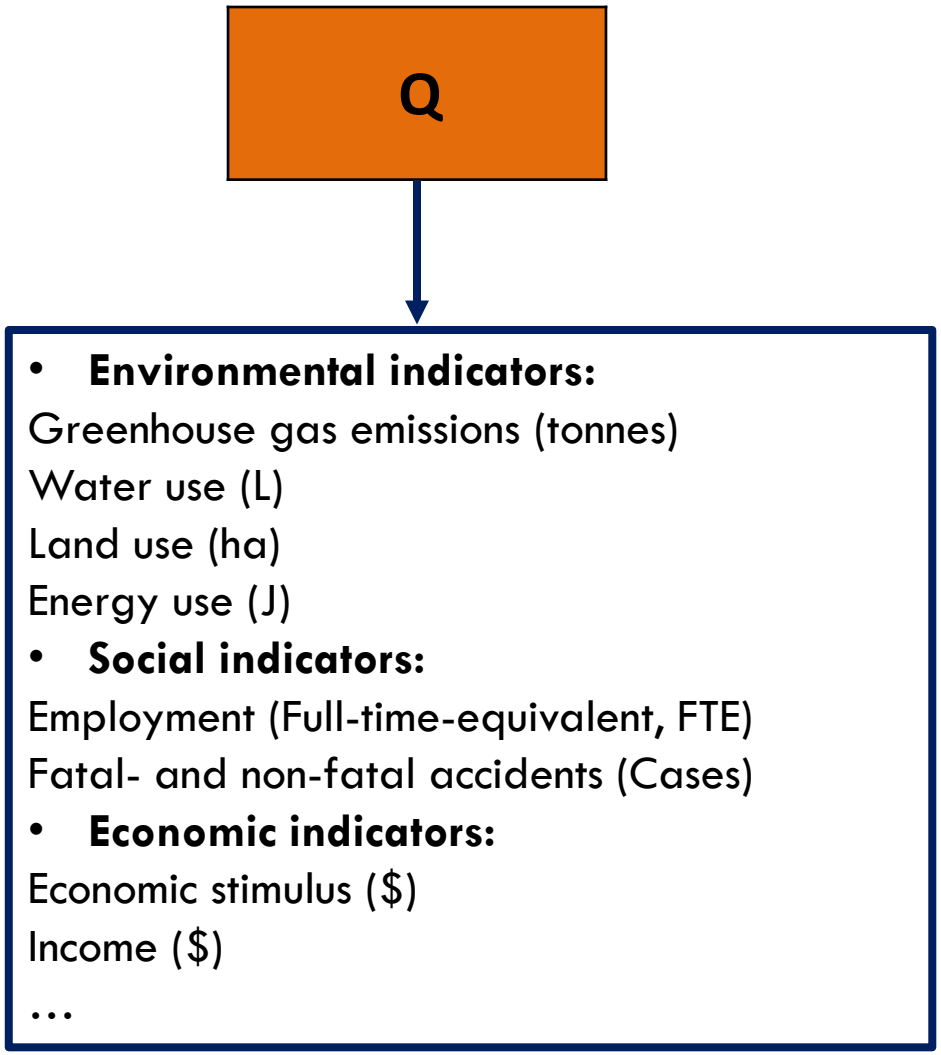
- Create an identity matrix (same size as the **A** matrix)
- Calculate **I – A**
- Calculate the inverse of **I – A** using MINVERSE

- **T** matrix; **A** matrix; **L** matrix – SAME SIZE!
- **A** matrix – Direct requirements matrix
- **L** matrix – Total requirements matrix

Recap – physical / satellite system



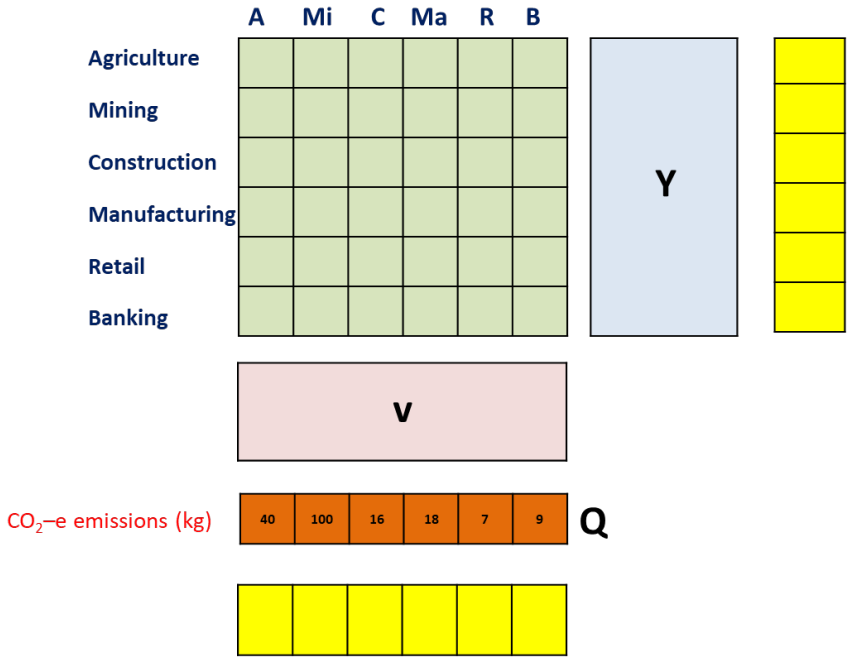
Satellite account matrix



Satellite account matrix



Satellite account matrix supports Production-based accounting



In this example (using GHG emissions as our environmental indicator) the **Q** matrix tells us how many kg of CO_2-e are emitted by each sector in order to produce its total output.

There is a direct connection between the sector and the GHG emitted as an input to production for that sector.

Production-based accounting: The Paris Agreement

Article 4

13. Parties shall account for their nationally determined contributions. In accounting for anthropogenic emissions and removals corresponding to their nationally determined contributions, Parties shall promote environmental integrity, transparency, accuracy, completeness, comparability and consistency, and ensure the avoidance of double counting, in accordance with guidance adopted by the Conference of the Parties serving as the meeting of the Parties to this Agreement.

United Nations 2015. Paris Agreement. *Article 4*.

Sometimes the term 'territorial emissions' is used

Production-based accounting: Is it sufficient?

(Production-based accounts ≈ Production-based footprint ≈ Territorial footprint)

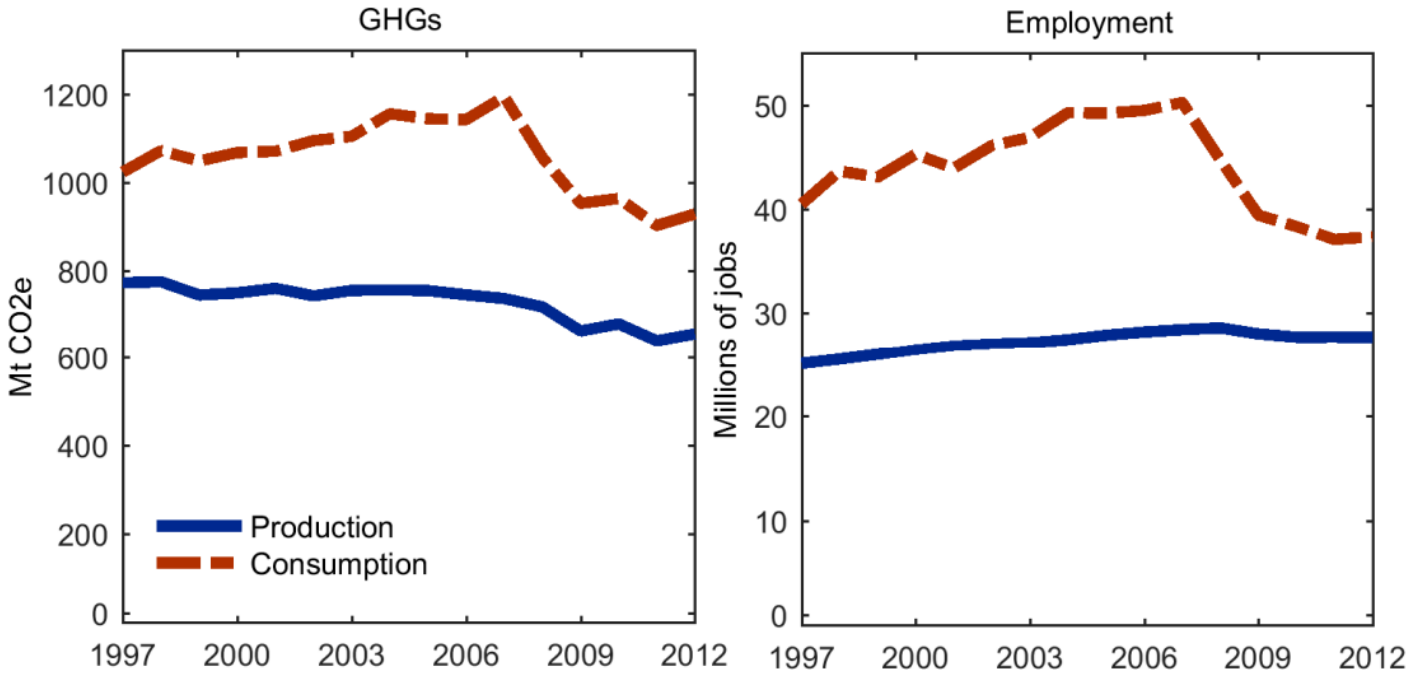
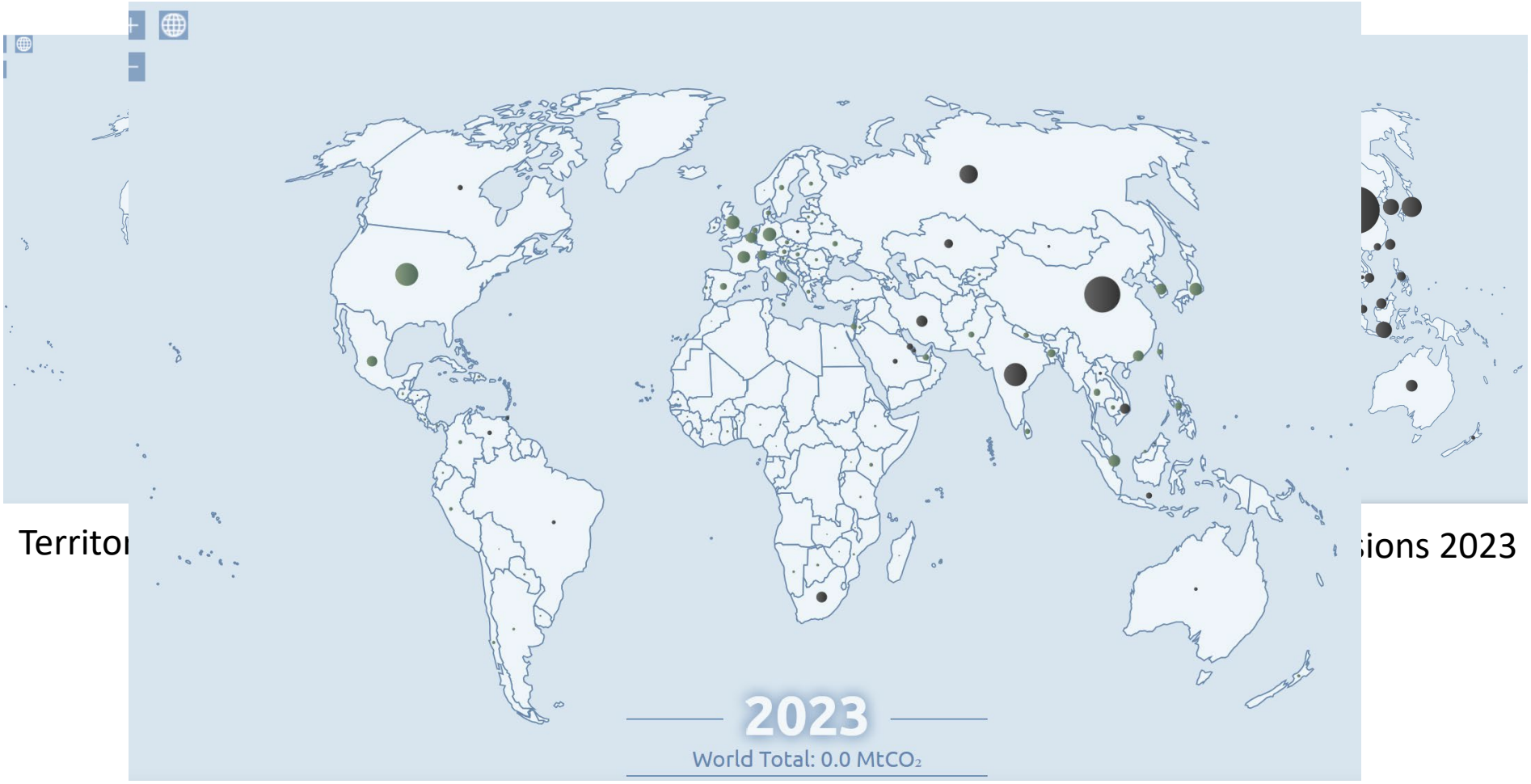


Figure 1. UK's greenhouse gas (GHGs) and employment historical trends (production and consumption).

Sakai, M., Owen, A. & Barrett, J. 2017. The UK's Emissions and Employment Footprints: Exploring the Trade-Offs. *Sustainability (Basel, Switzerland)*, 9, 1242.

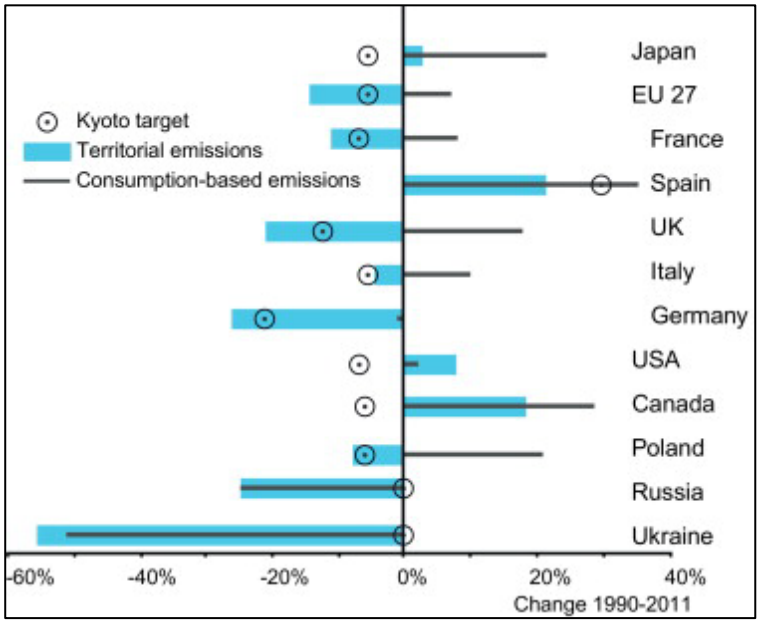
Production-based accounting: Is it sufficient?



Territor ions 2023

Global Carbon Atlas: https://globalcarbonatlas.org/emissions/carbon-emissions/

Consumption-based accounting

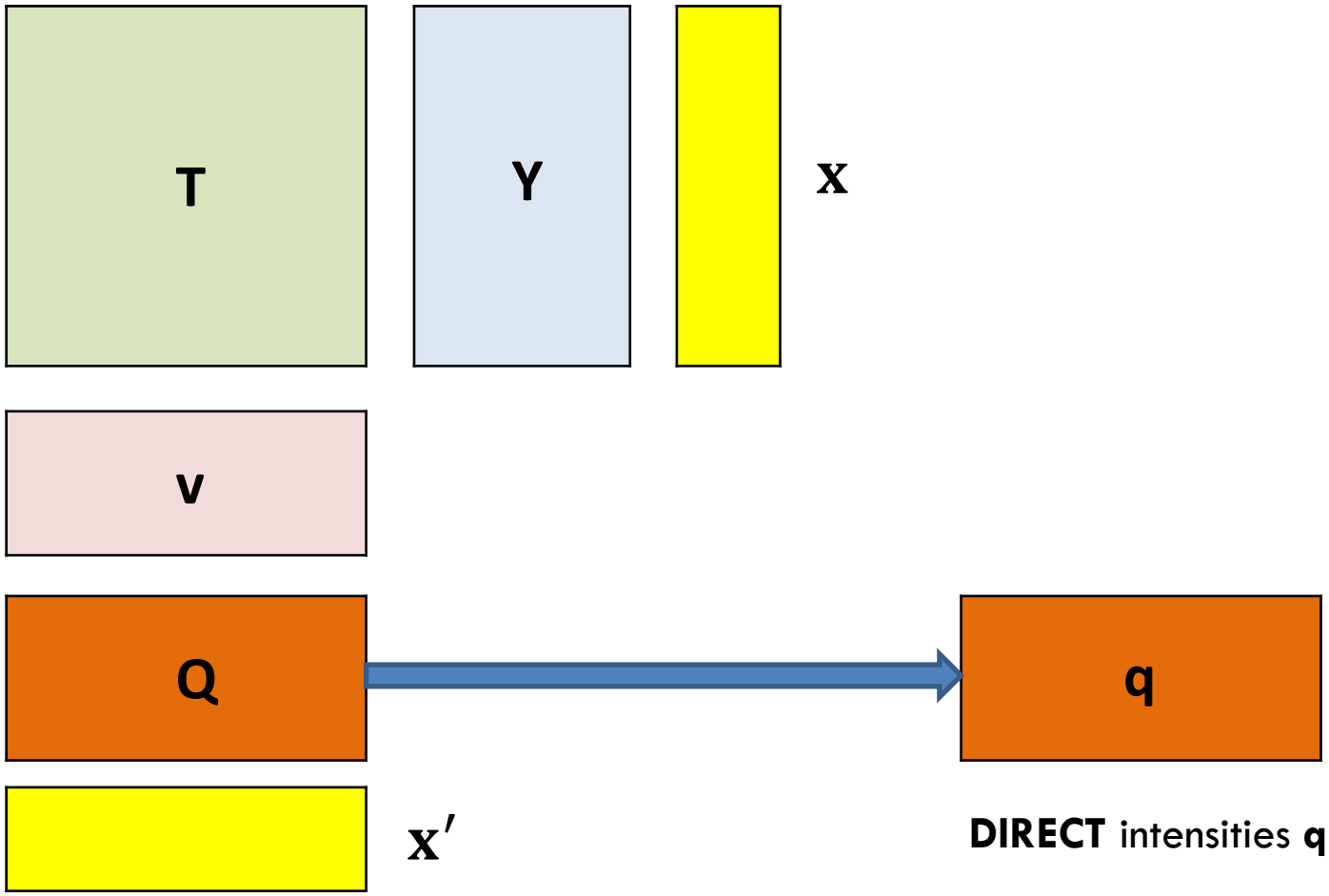


Kanemoto, K, Moran, D, Lenzen, M & Geschke, A 2014, 'International trade undermines national emission reduction targets: New evidence from air pollution', *Global environmental change*, vol. 24, no. 1, pp. 52-59.

In order to connect the GHG emissions to the point of consumption, we need to find a way to connect the data in the **Q** matrix to final demand.

Calculate the consumption footprint

Footprint calculations: intensities

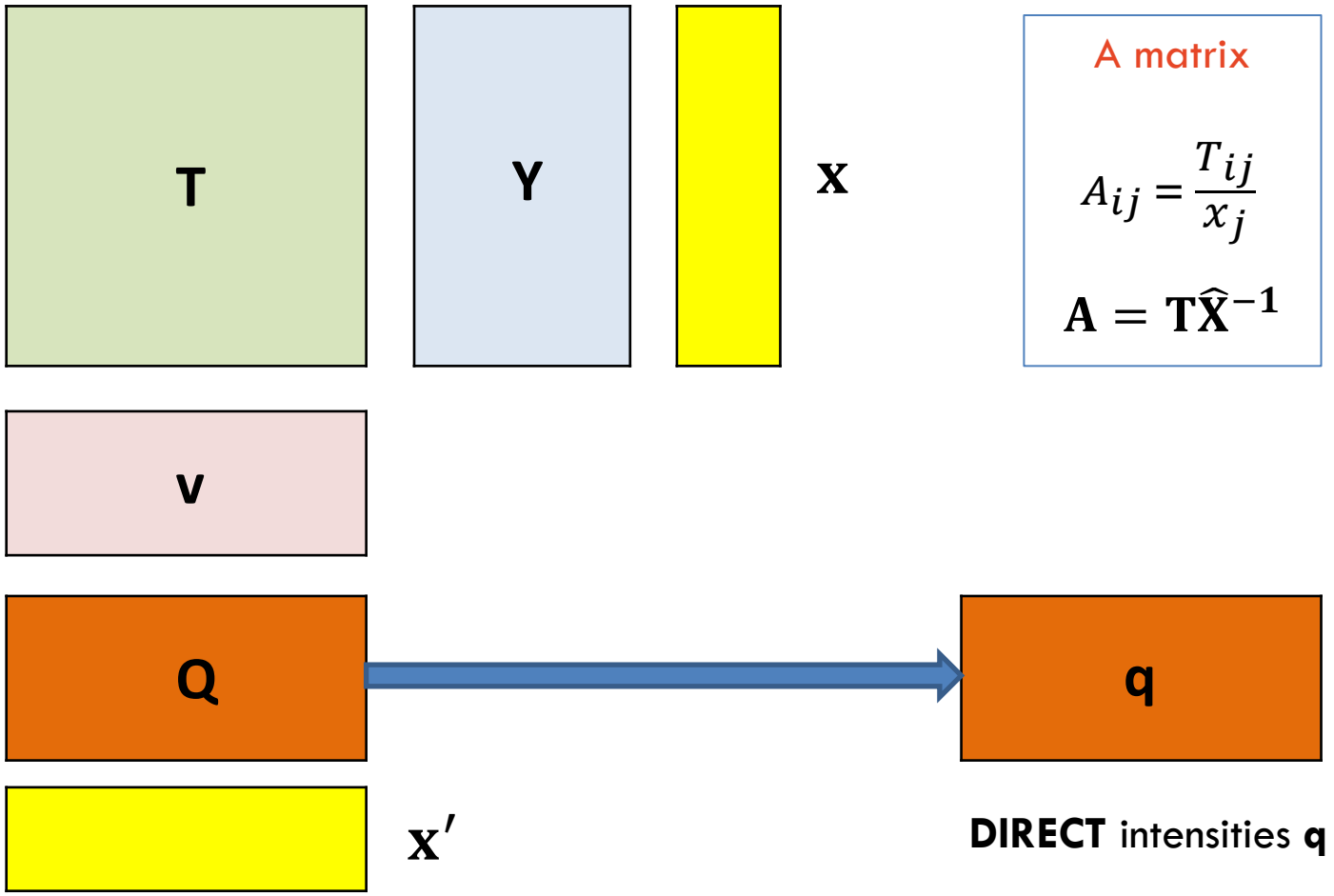


DIRECT intensities **q**

Direct impacts per dollar of total output *

* Note that we are using $ as the monetary unit for simplicity. If the IO table used a different monetary unit, such as Euro, Yen, Ringgit, Rupiah etc, then this would read differently.

Footprint calculations: intensities



A matrix

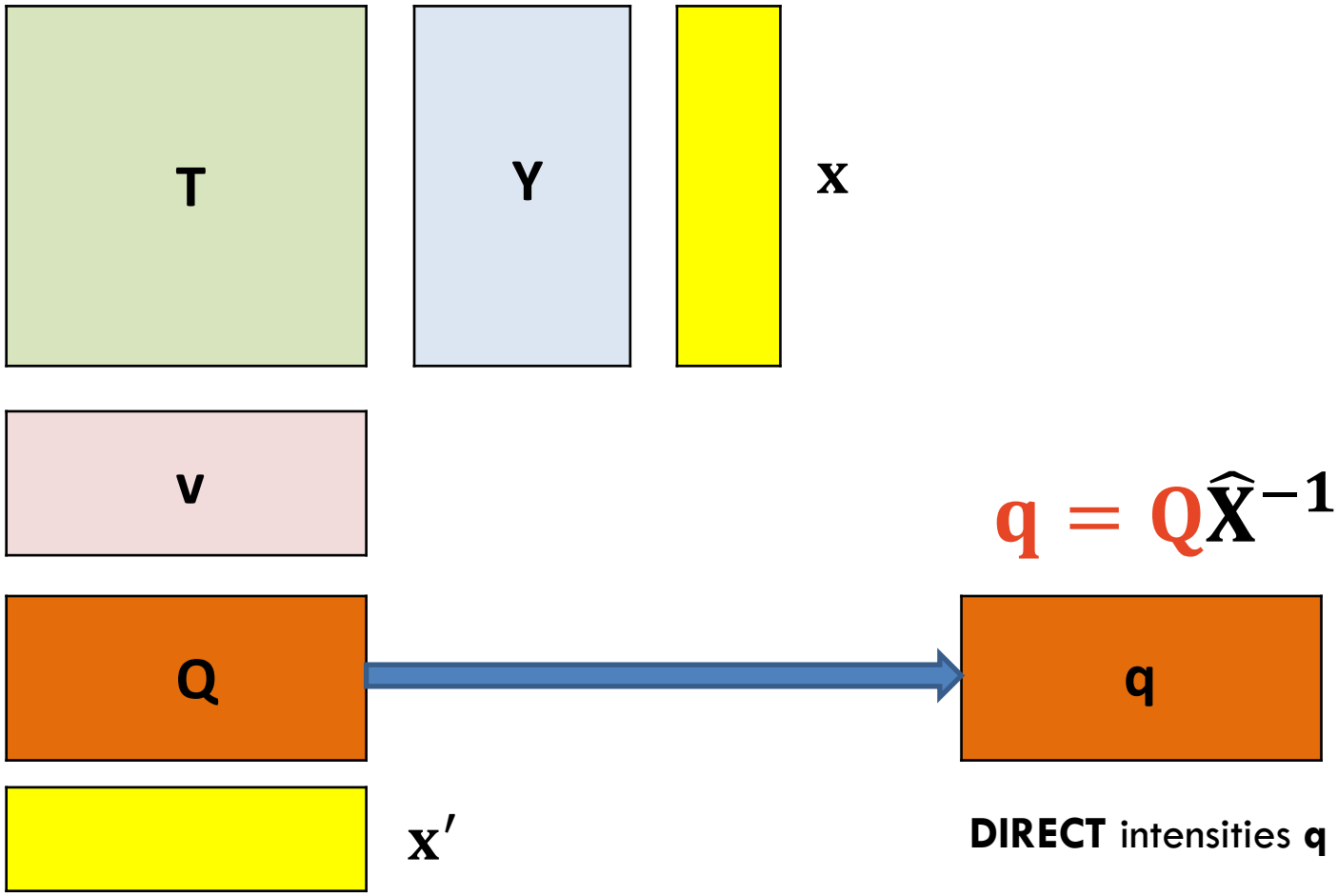
$$A_{ij} = \frac{T_{ij}}{x_j}$$

$$\mathbf{A} = \mathbf{T}\hat{\mathbf{X}}^{-1}$$

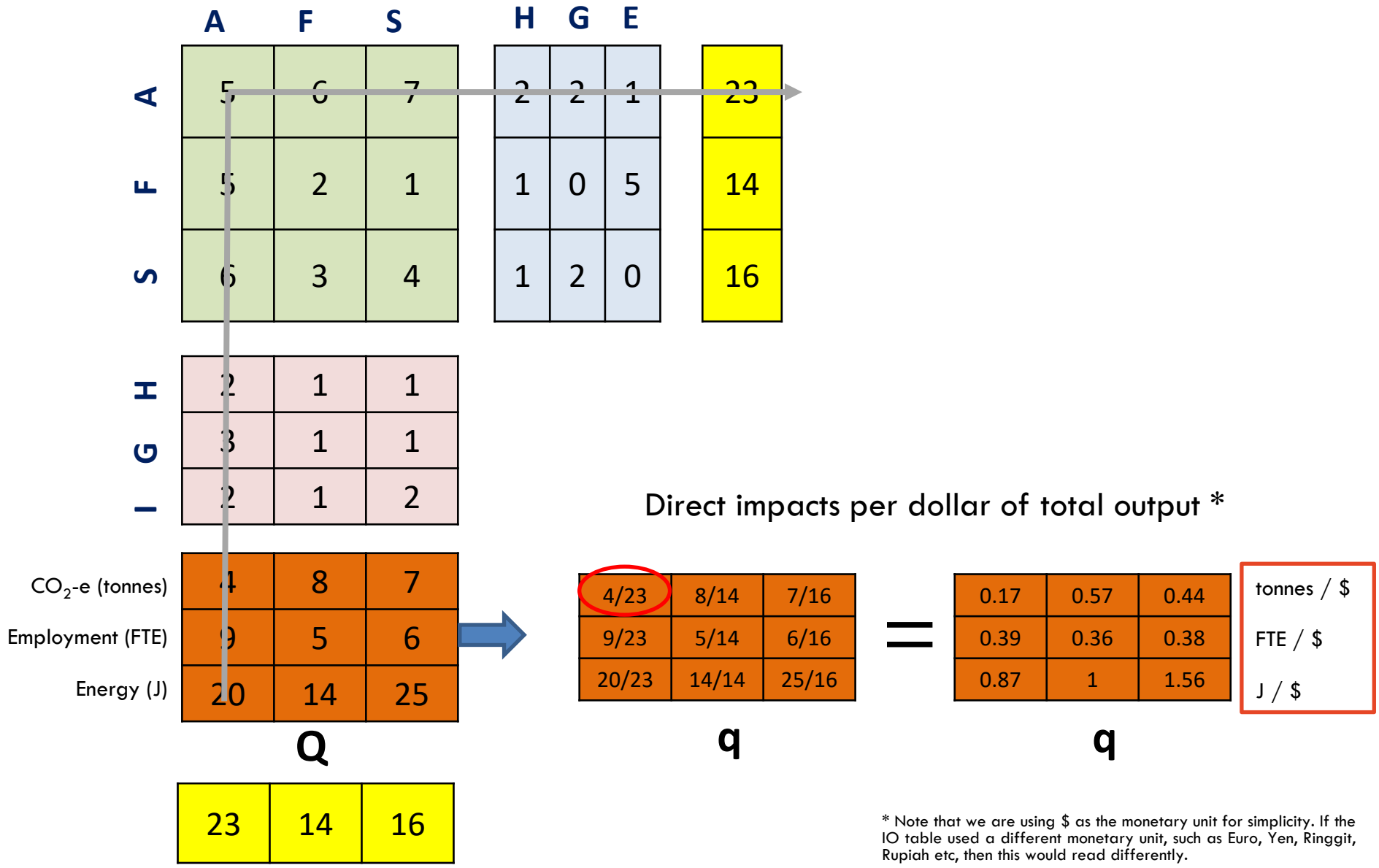
DIRECT intensities **q**

Similar calculation to the one used to calculate the **DIRECT** requirements (**A**) matrix

Footprint calculations: intensities



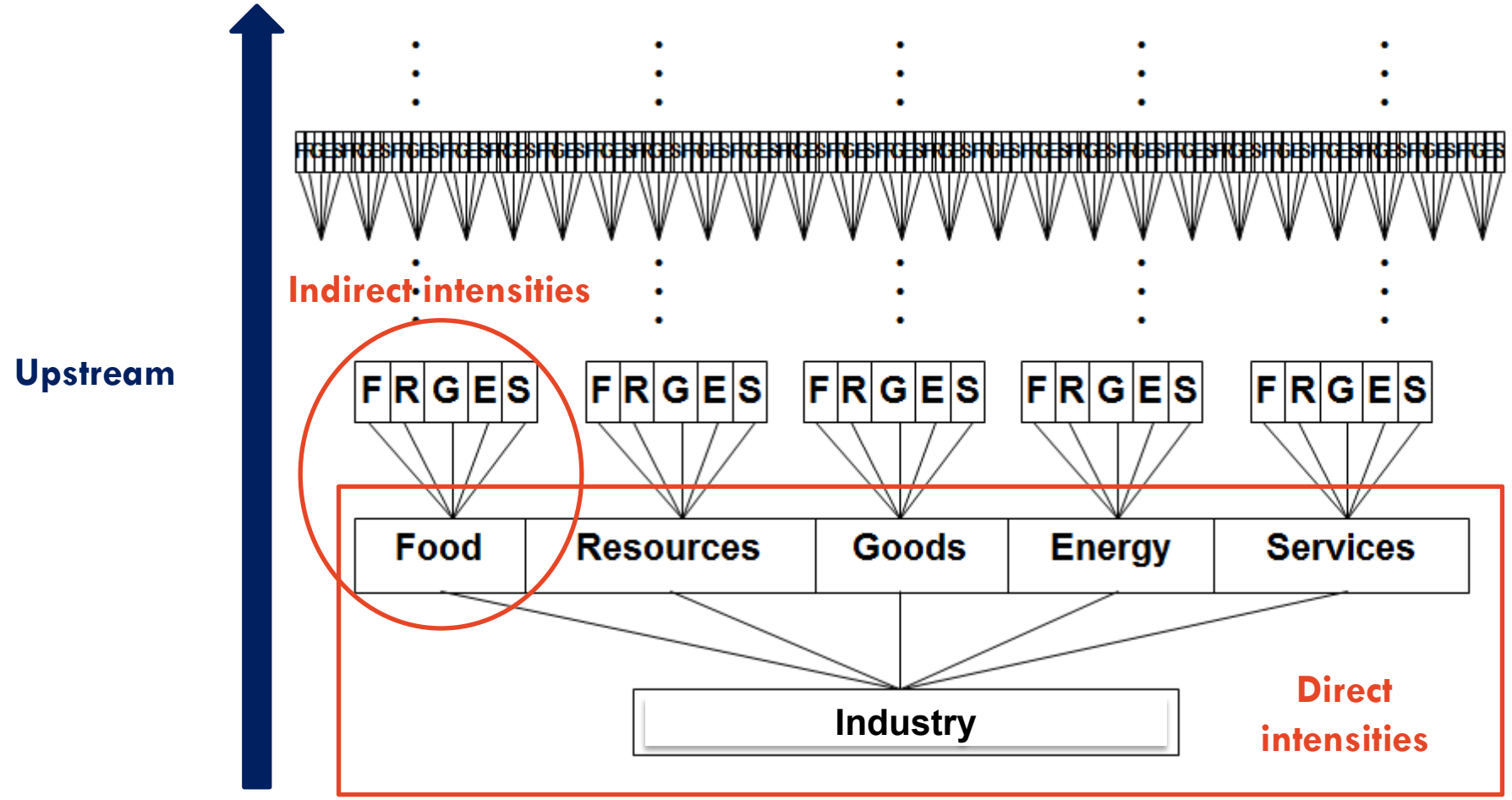
Footprint calculations: intensities



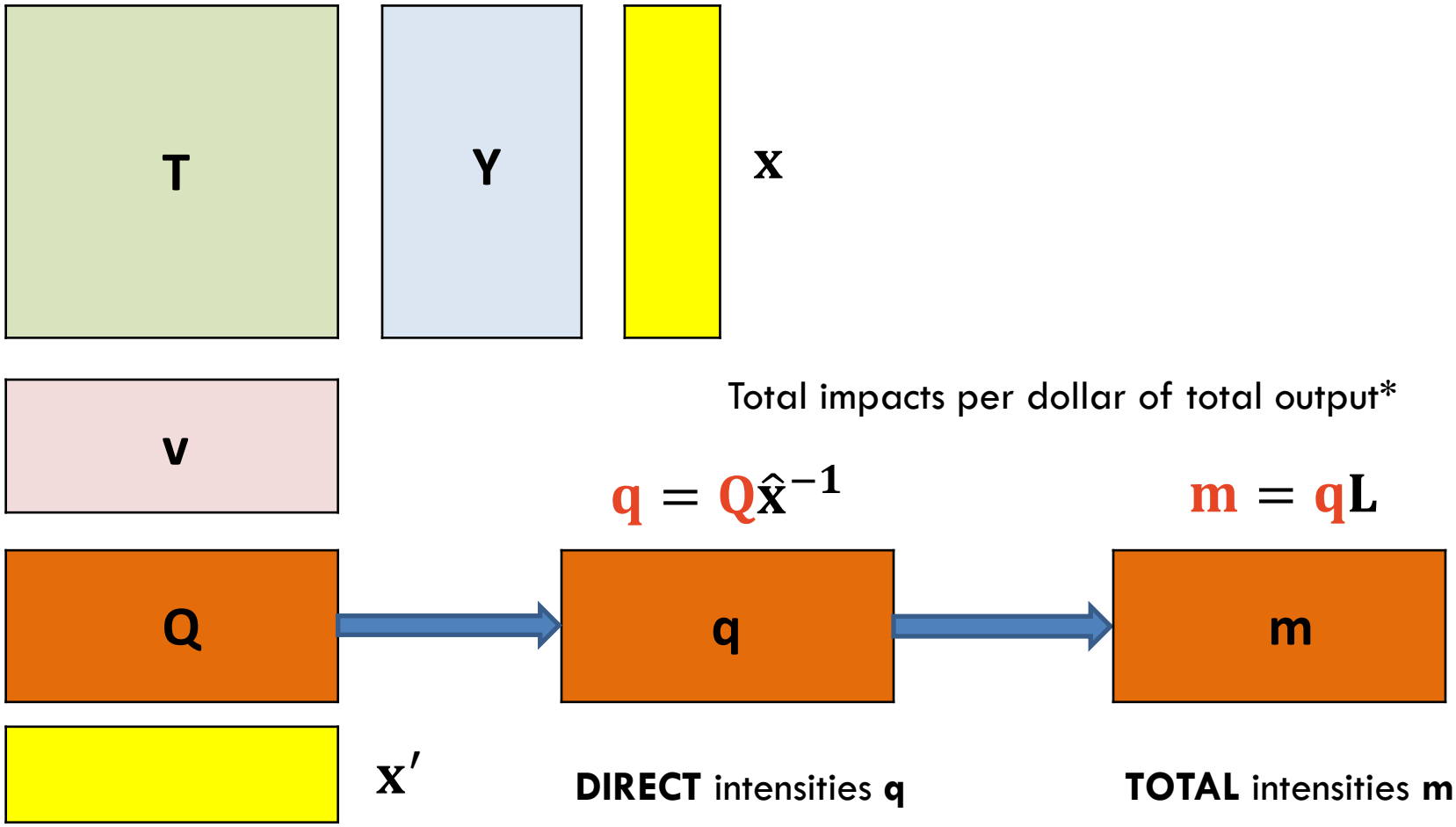
* Note that we are using $ as the monetary unit for simplicity. If the IO table used a different monetary unit, such as Euro, Yen, Ringgit, Rupiah etc, then this would read differently.

Do industries only rely on direct inputs?

We need to consider the supply chain!



Footprint calculations: intensities



* Note that we are using $ as the monetary unit for simplicity. If the IO table used a different monetary unit, such as Euro, Yen, Ringgit, Rupiah etc, then this would read differently.

Footprints calculations: bringing it all together !

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1} y$$

$$\mathbf{x} = \mathbf{L} y$$

Multiply both sides by **q**

$$\mathbf{q}\mathbf{x} = \mathbf{q}\mathbf{L} y$$

$$\mathbf{q}\mathbf{x} = \mathbf{m} y$$

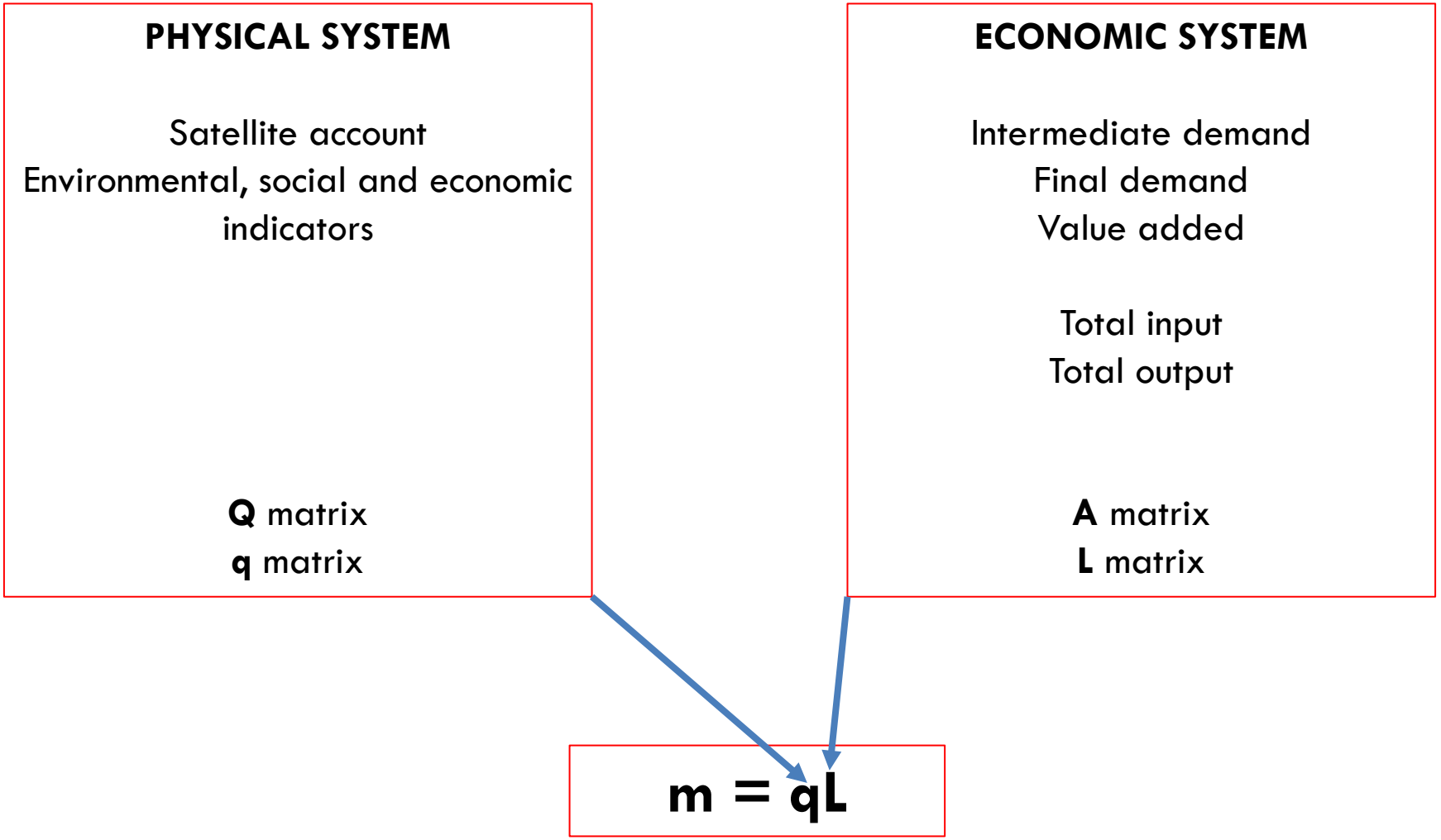
Direct intensities

$$Footprint = \mathbf{m} y = \mathbf{q}\mathbf{L} y$$

Total intensities

Units for the footprint will be the same as the indicator that it is being calculated for

Footprint calculations: intensities



Footprint calculations: bringing it all together !

$$\mathbf{qx} = \mathbf{m}\boldsymbol{y}$$

$$\mathbf{Q} = \mathbf{m}\boldsymbol{y}$$

Remember
$q = Q\widehat{X}^{-1}$

Producer perspective

Q and **q** capture direct impacts at the point of production for total output **x**

Units for the footprint will be the same as for the indicator that it is being calculated for

Footprint calculations: bringing it all together !

$$\mathbf{q}\mathbf{x} = \mathbf{m}\boldsymbol{y}$$

$$\mathbf{Q} = \mathbf{m}\boldsymbol{y}$$

Consumer perspective

y is the driver of those impacts
L captures all supply chains

The economic transactions along those supply chains connect the final demand at **y** to the direct impact **Q**

Footprint calculations: bringing it all together !

$$\mathbf{qx} = \mathbf{m}\boldsymbol{y}$$

$$\mathbf{Q} = \mathbf{m}\boldsymbol{y}$$

The sum of the production-based footprint (**Q**) across all sectors will be equal to the sum of the consumption-based footprint (**qLy**) across all sectors when y = total final demand

Who uses these techniques?

Academia

LETTER

doi:10.1038/nature11145

International trade drives biodiversity threats in developing nations

M. Lenzen[1], D. Moran[1], K. Kanemoto[1,2], B. Foran[1,3], L. Lobefaro[1,4] & A. Geschke[1]

Economic damage and spillovers from a tropical cyclone

Manfred Lenzen[1], Arunima Malik[1,2], Steven Kenway[3], Peter Daniels[4], Ka Leung Lam[3], and Arne Geschke[1]

[1]ISA, School of Physics A28, The University of Sydney, NSW, 2006, Sydney, Australia
[2]Discipline of Accounting, The University of Sydney Business School, The University of Sydney, NSW, 2006, Sydney, Australia
[3]School of Chemical Engineering, The University of Queensland, St Lucia, 4072, Australia
[4]School of Environment, Griffith University, Brisbane, 4222, Australia

The carbon footprint of Australian health care

Arunima Malik, Manfred Lenzen, Scott McAlister, Forbes McGain

Summary

Background Carbon footprints stemming from health care have been found to be variable, from 3% of the total national CO_2 equivalent (CO_2e) emissions in England to 10% of the national CO_2e emissions in the USA. We aimed to measure the carbon footprint of Australia's health-care system.

Methods We did an observational economic input–output lifecycle assessment of Australia's health-care system. All expenditure data were obtained from the 15 sectors of the Australian Institute of Health and Welfare for the financial year 2014–15. The Australian Industrial Ecology Virtual Laboratory (IELab) data were used to obtain CO_2e emissions per AUS$ spent on health care.

Findings In 2014–15 Australia spent $161·6 billion on health care that led to CO_2e emissions of about 35 772 (68% CI 25 398–46 146) kilotonnes. Australia's total CO_2e emissions in 2014–15 were 494 930 kilotonnes, thus health care represented 35 772 (7%) of 494 930 kilotonnes total CO_2e emissions in Australia. The five most important sectors within health care in decreasing order of total CO_2e emissions were: public hospitals (12 295 [34%] of 35 772 kilotonnes CO_2e), private hospitals (3635 kilotonnes [10%]), other medications (3347 kilotonnes [9%]), benefit-paid drugs (3257 kilotonnes [9%]), and capital expenditure for buildings (2776 kilotonnes [8%]).

Interpretation The carbon footprint attributed to health care was 7% of Australia's total; with hospitals and pharmaceuticals the major contributors. We quantified Australian carbon footprint attributed to health care and identified health-care sectors that could be ameliorated. Our results suggest the need for carbon-efficient procedures, including greater public health measures, to lower the impact of health-care services on the environment.

Lancet Planet Health 2018; 2: e27–35

See Comment page e8

Integrated Sustainability Analysis, School of Physics (A Malik PhD, Prof M Lenzen PhD), **and School of Public Health** (F McGain PhD), **University of Sydney, Sydney, NSW, Australia; Ecoquantum Life Cycle Assessment, Melbourne, VIC, Australia** (S McAlister MEngSci); **and Departments of Anaesthesia and Intensive Care, Western Health, Footscray, VIC, Australia** (F McGain)

Correspondence to Dr Forbes McGain, Departments of Anaesthesia and Intensive Care, Western Health, Footscray, VIC 3011, Australia
forbes.mcgain@wh.org.au

The material footprint of nations

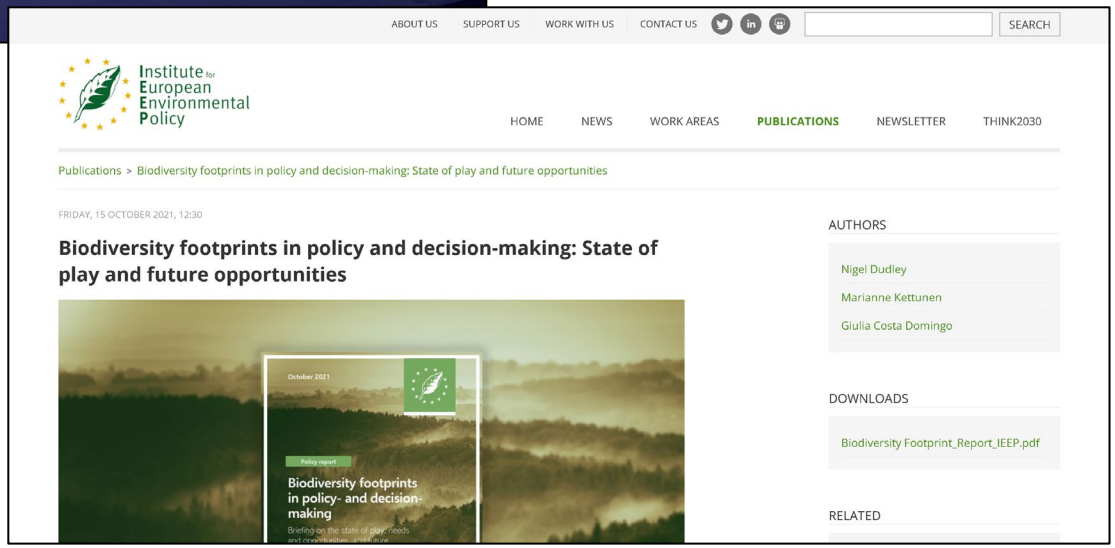
Thomas O. Wiedmann[a,b,c,1], Heinz Schandl[b,d], Manfred Lenzen[c], Daniel Moran[c,e], Sangwon Suh[f], James West[b], and Keiichiro Kanemoto[c,g]

Who uses these techniques?

Government – policy making



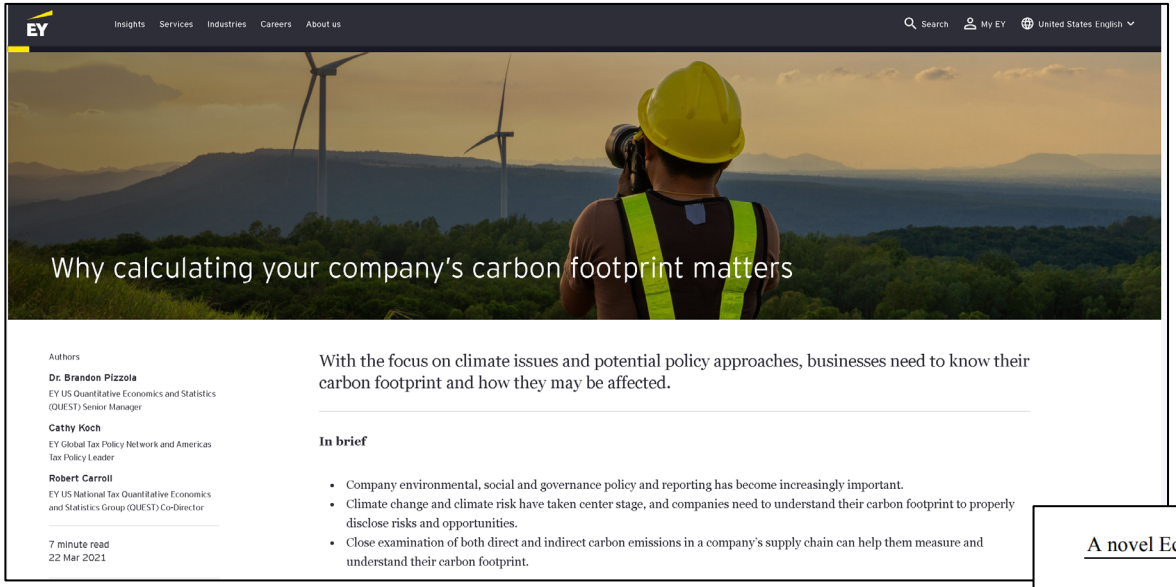
https://www.unsdsn.org/spillover-effects



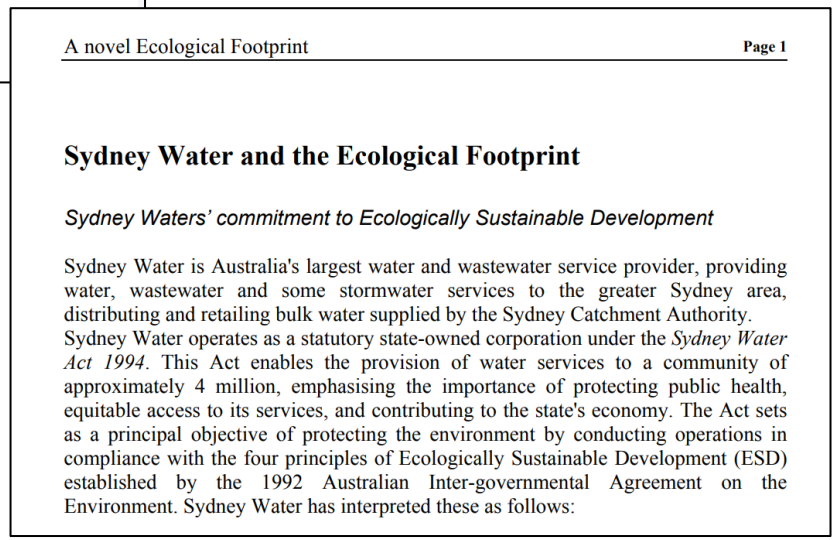
https://ieep.eu/publications/biodiversity-footprints-in-policy-and-decision-making-state-of-play-and-future-opportunities

Who uses these techniques?

Industry



https://www.ey.com/en_us/tax/why-calculating-your-company-s-carbon-footprint-matters



https://isa.org.usyd.edu.au/publications/documents/SydneyWaterFootprint.pdf

Example footprint Application: Biodiversity loss

Step 1: Construct the **Q** matrix, but . . . How do you measure biodiversity?

"the variability among living organisms from all sources including, *inter alia*, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part: this includes diversity within species, between species and of ecosystems"

Convention on Biological Diversity (CBD)
https://www.cbd.int/intro/

Example footprint Application: Biodiversity loss

Step 1: Construct the **Q** matrix, but . . . How do you measure biodiversity? Potential options:

- Number of species
- Number of individuals within each species
- Number of species in each ecosystem
- Number of ecosystems
- Genetic variability
- Genome sequencing

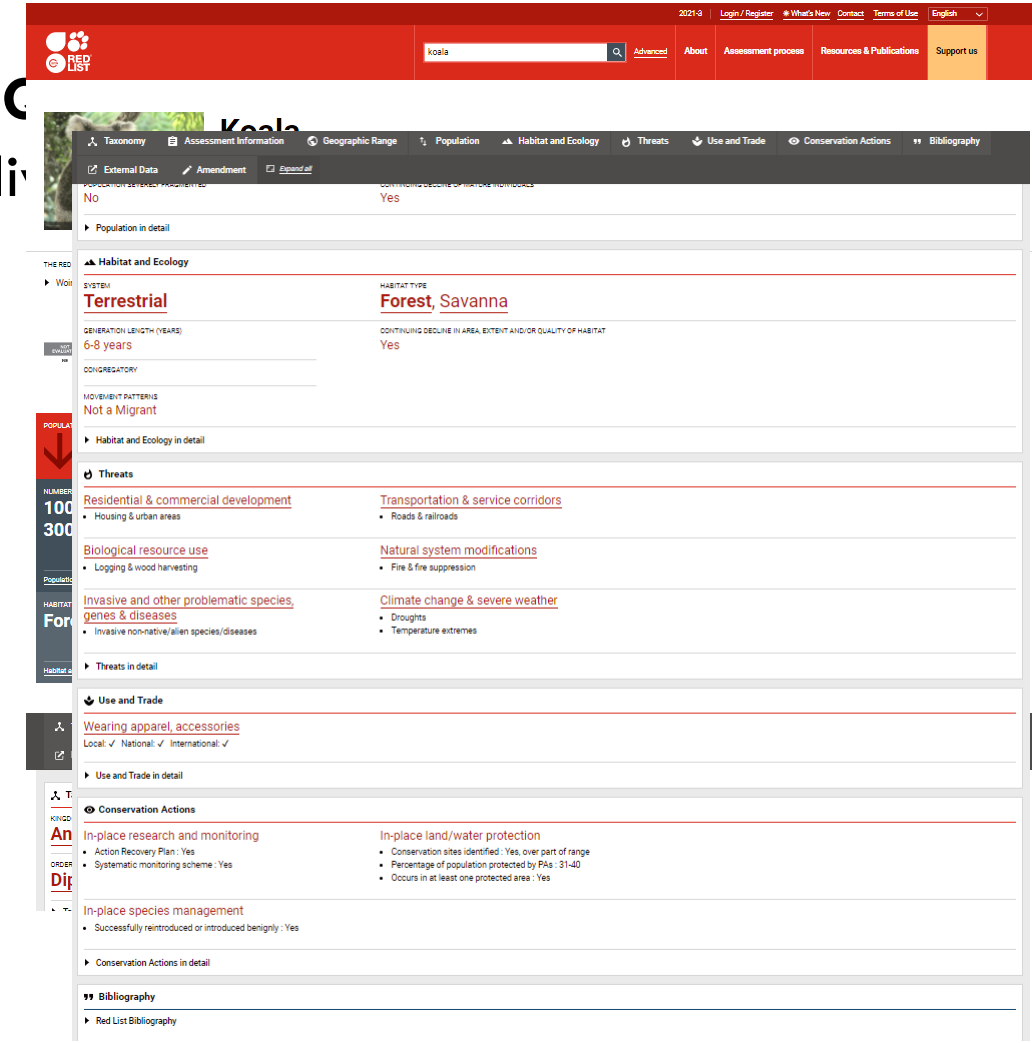
Example footprint Application: Biodiversity loss

Step 1: Construct the C… measure biodi…

Choose the extinction risk of species as a proxy for measuring biodiversity loss.

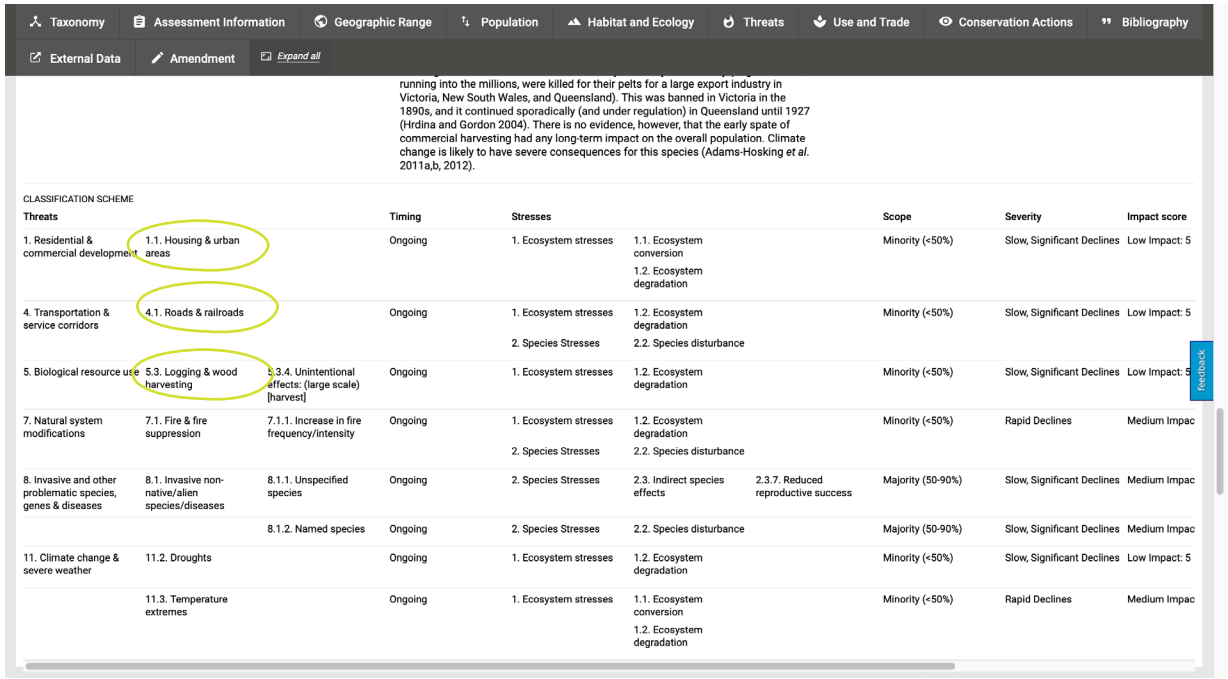
The IUCN Red List of Threatened Species provides detailed information on each assessed species:

- Its extinction risk category (Critically Endangered, Endangered, Vulnerable, Near Threatened, Least Concern)
- Its distribution range
- Its habitat and elevation preferences
- The threats acting on it



Example footprint Application: Biodiversity loss

Step 1: Construct the **Q** matrix, but . . . How do you connect this information to economic activity so we can use IO methodology?



Taxonomy | Assessment Information | Geographic Range | Population | Habitat and Ecology | Threats | Use and Trade | Conservation Actions | Bibliography

External Data | Amendment | Expand all

running into the millions, were killed for their pelts for a large export industry in Victoria, New South Wales, and Queensland). This was banned in Victoria in the 1890s, and it continued sporadically (and under regulation) in Queensland until 1927 (Hrdina and Gordon 2004). There is no evidence, however, that the early spate of commercial harvesting had any long-term impact on the overall population. Climate change is likely to have severe consequences for this species (Adams-Hosking *et al.* 2011a,b, 2012).

CLASSIFICATION SCHEME

Threats			Timing	Stresses			Scope	Severity	Impact score
1. Residential & commercial development	1.1. Housing & urban areas		Ongoing	1. Ecosystem stresses	1.1. Ecosystem conversion 1.2. Ecosystem degradation		Minority (<50%)	Slow, Significant Declines	Low Impact: 5
4. Transportation & service corridors	4.1. Roads & railroads		Ongoing	1. Ecosystem stresses 2. Species Stresses	1.2. Ecosystem degradation 2.2. Species disturbance		Minority (<50%)	Slow, Significant Declines	Low Impact: 5
5. Biological resource use	5.3. Logging & wood harvesting	5.3.4. Unintentional effects: (large scale) [harvest]	Ongoing	1. Ecosystem stresses	1.2. Ecosystem degradation		Minority (<50%)	Slow, Significant Declines	Low Impact: 5
7. Natural system modifications	7.1. Fire & fire suppression	7.1.1. Increase in fire frequency/intensity	Ongoing	1. Ecosystem stresses 2. Species Stresses	1.2. Ecosystem degradation 2.2. Species disturbance		Minority (<50%)	Rapid Declines	Medium Impac
8. Invasive and other problematic species, genes & diseases	8.1. Invasive non-native/alien species/diseases	8.1.1. Unspecified species	Ongoing	2. Species Stresses	2.3. Indirect species effects	2.3.7. Reduced reproductive success	Majority (50-90%)	Slow, Significant Declines	Medium Impac
		8.1.2. Named species	Ongoing	2. Species Stresses	2.2. Species disturbance		Majority (50-90%)	Slow, Significant Declines	Medium Impac
11. Climate change & severe weather	11.2. Droughts		Ongoing	1. Ecosystem stresses	1.2. Ecosystem degradation		Minority (<50%)	Slow, Significant Declines	Low Impact: 5
	11.3. Temperature extremes		Ongoing	1. Ecosystem stresses	1.1. Ecosystem conversion 1.2. Ecosystem degradation		Minority (<50%)	Rapid Declines	Medium Impac

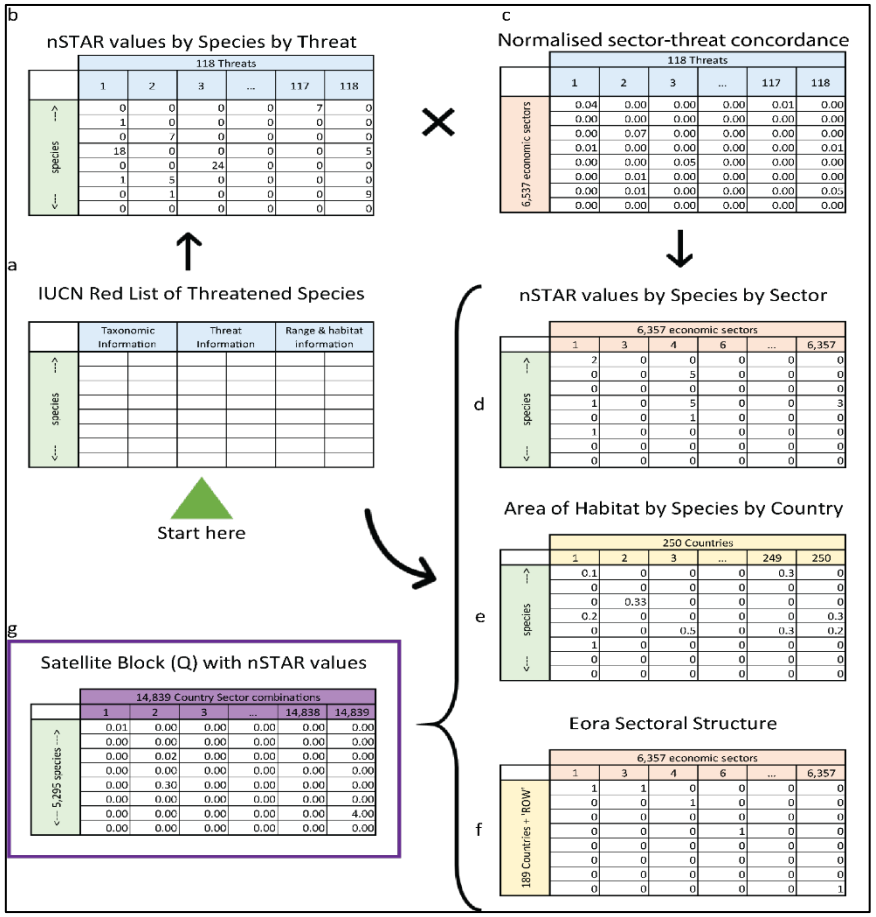
Feedback

Connected to economic activity

Threat impact score	Negligible or no decline	Slow, significant declines	Rapid declines	Very rapid declines
Minority (<50%)	0	5	7	24
Majority (50-90%)	0	9	18	52
Whole (>90%)	1	10	24	63

Example footprint Application: Biodiversity loss

Step 1: Construct the **Q** matrix, lots of matrix manipulation



Example footprint Application: Biodiversity loss

Step 1: Construct the **Q** matrix, calculate the production-based account of extinction-risk in Australia

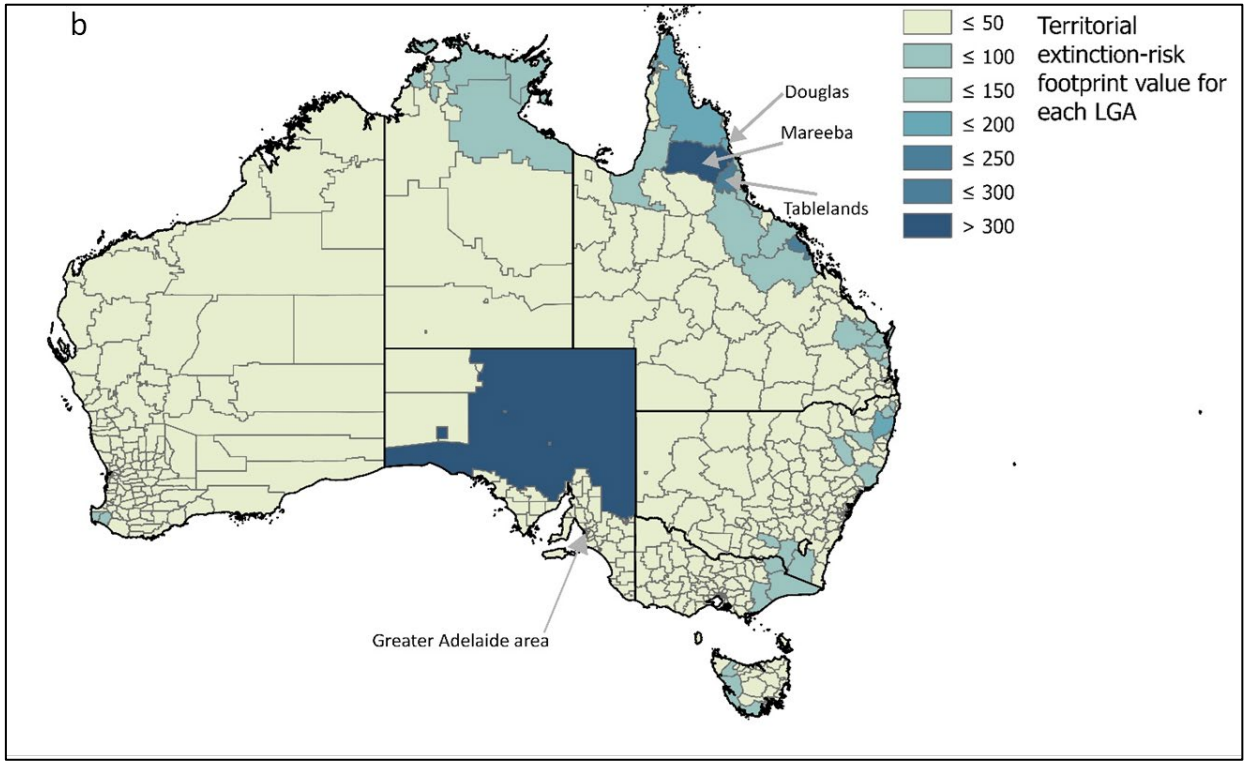
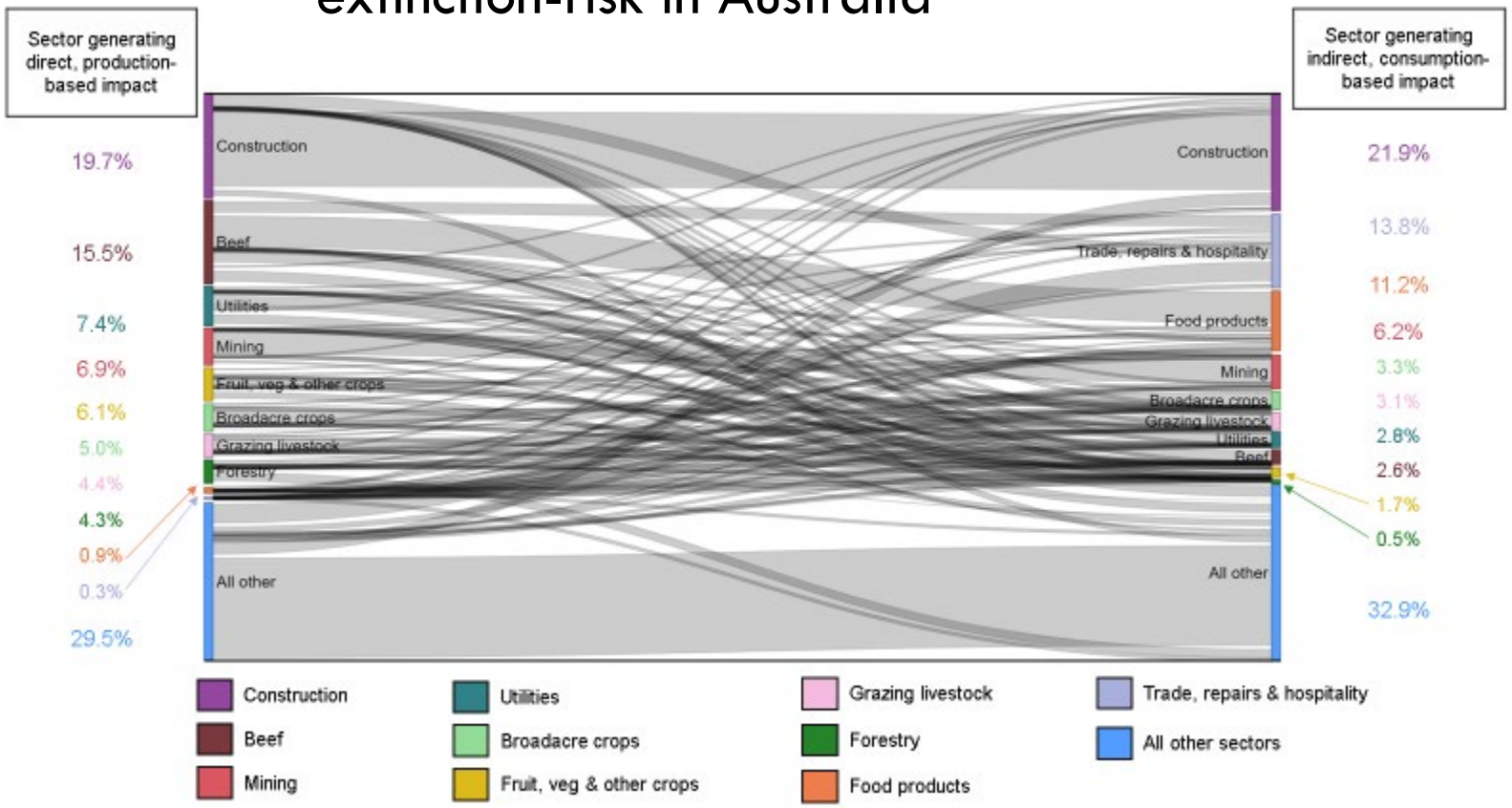


Figure 1b. The distribution of territorial extinction-risk footprint across Australia. Local government areas are colour-coded according the size of the territorial extinction-risk footprint, a measure of extinction risk which takes into account a species' extinction risk category and the scope and severity of each threat acting on it. The darker the colour, the higher the relevant value. Note that there are no units for extinction-risk footprint.

Irwin, A & Geschke, A, 2023, 'A consumption-based analysis of extinction risk in Australia', Conservation Letters, vol. 16, no. 3, doi: 10.1111/conl.12942.

Example footprint Application: Biodiversity loss

Step 2: Calculate the consumption-based account of extinction-risk in Australia



Irwin, A & Geschke, A, 2023, 'A consumption-based analysis of extinction risk in Australia', Conservation Letters, vol. 16, no. 3, doi: 10.1111/conl.12942.

Example footprint Application: Biodiversity loss

Step 2: Calculate the consumption-based account of extinction-risk in Australia

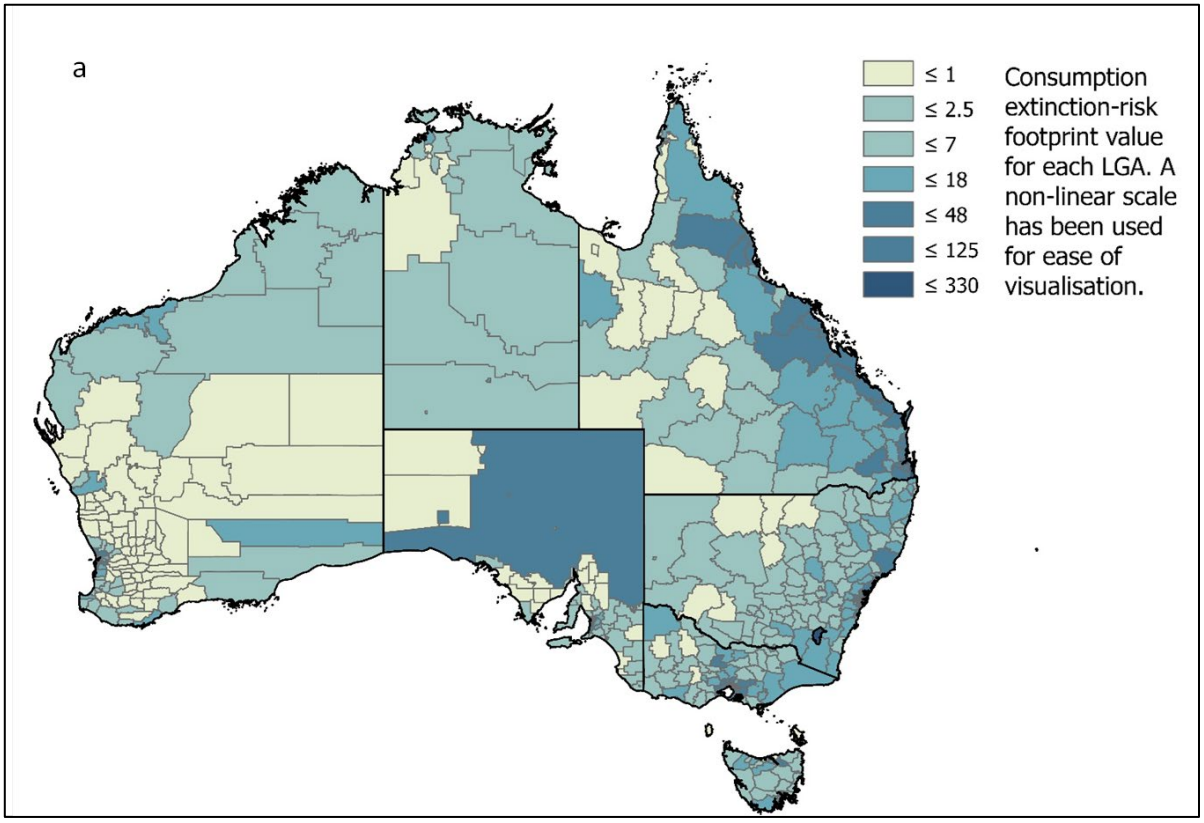


Figure 4a. The distribution of consumption extinction-risk footprint across Australia. Local government areas are colour-coded according to the size of the consumption extinction-risk footprint. The darker the colour, the higher the value.

Irwin, A & Geschke, A, 2023, 'A consumption-based analysis of extinction risk in Australia', Conservation Letters, vol. 16, no. 3, doi: 10.1111/conl.12942.

Footprints calculations: bringing it all together !

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1} y$$

$$\mathbf{x} = \mathbf{L} y$$

Multiply both sides by **q**

$$\mathbf{q}\mathbf{x} = \mathbf{q}\mathbf{L} y$$

$$\mathbf{q}\mathbf{x} = \mathbf{m} y$$

$$Footprint = \mathbf{m} y = \mathbf{q}\mathbf{L} y$$

Units for the footprint will be the same as the indicator that it is being calculated for

Tutorial



THE UNIVERSITY OF
SYDNEY

Tutorial Activity

In week 5, we constructed a satellite block for the following input-output matrix. The satellite block had four indicators – CO_2 emissions, energy, water and nitrogen.

($100million)	Sheep, Grain, Beef and Dairy Cattle	Poultry and Other Livestock	Coal Mining	Beer Manufacturing	Textile Manufacturing	Chemical Manufacturing	Retail Trade	Techinical Services	Banking	Defence	Households	Government	Exports
Sheep, Grain, Beef and Dair Cattle	40	30	1	40	2	4	1	1	0	0	3	10	72
Poultry and Other Livestock	35	25	0	2	10	10	5	1	0	0	20	5	48
Coal Mining	2	1	25	0	0	10	0	15	0	0	0	12	125
Beer Manufacturing	1	2	5	38	0	2	10	0	0	0	14	10	59
Textile Manufacturing	10	5	0	5	25	12	3	1	0	5	10	6	28
Chemical Manufacturing	22	25	42	8	9	24	1	2	0	32	5	15	15
Retail Trade	12	8	4	3	12	4	24	28	21	8	32	8	12
Techinical Services	24	25	19	12	20	30	8	20	48	50	15	20	7
Banking	8	7	25	6	4	12	15	25	30	50	28	40	44
Defence	0	0	0	0	0	0	0	0	0	12	0	146	100
Households	12	3	15	5	6	10	55	85	63	10			
Government	10	9	13	4	5	10	15	12	20	52			
Imports	28	21	41	18	17	72	39	108	112	39			

For the satellite block, complete the following:

1. **Calculate q and m.**
   Use the formulae presented in the lecture to calculate q and m.

2. **Calculate the footprints for the entire final demand of your economy**
   Work out the total final demand vector for this economy and calculate the footprints associated with total final demand. Compare this to your indicator totals. Are they the same?

3. **Calculate the footprints for the first column of final demand (Households)**

4. **Calculate the footprints for the second column of final demand (Government)**

5. **Calculate the footprints for the third column of final demand (Exports)**

6. **Add up the footprints calculated in steps 3-5 and compare them to the total footprints calculated in step 2? Are they the same?**

Homework for Week 6

Homework week 6 (Due: October 28, 2026)

Use your 14-sector input-output table, including the satellite matrix (Q) that you created in week 5, to complete the following:

1. **Calculate q and m.**

Use the formulae presented in the lecture to calculate q and m.

2. **Calculate the footprints for the entire final demand of your economy**

Use the total final demand vector for your economy to calculate the footprint associated with total final demand for each of your indicators. Compare this to the indicator totals. Are they the same?

3. **Calculate the footprint associated with the first column of final demand for each indicator**
4. **Calculate the footprint associated with the second column of final demand for each indicator**
5. **Calculate the footprint associated with the third column of final demand for each indicator**
6. **Add up the footprints calculated at steps 3), 4), and 5) and compare them to the total footprints calculated in step 2)? Are these two numbers the same for each indicator?**

Note: footprints need to be calculated for all 3 of the indicators in the satellite matrix you created in Week 5. Don't forget to include labels and units.

EXPLANATION: Explain the concept of direct intensities and total intensities in words (100 words). Why is your footprint answer in Q6 exactly the same as the answer in Q2 (100 words)?

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Key takeaways

- The **Q** block forms the basis for a production-based account of the indicator in question

- The **q** block is used, along with **L** and **y**, to calculate a consumption-based account of the indicator in questions

$$\boldsymbol{Footprint = \mathbf{m}y = \mathbf{q}\mathbf{L}y}$$

Next week:

- Multi-region input output tables